

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 8 Summer 2025 (Regular)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTMEC801F - NON-CONVENTIONAL ENERGY RESOURCES

Time : 1.5 hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

1 An artificial pacemaker of tens of microwatt is made up of _____.

1. Li-ion battery
2. Secondary battery
3. Primary battery
4. All of the Above

The region of the atmosphere where human activity has the most impact is.

1. Troposphere
2. Stratosphere
3. Mesosphere
4. Ionosphere

3 A normal human being cannot see the things in the complete darkness because eyes see the objects only through _____.

1. Reflected visible light
2. Reflected IR
3. Inherent IR of the object being seen
4. Reflected UV

Flat plate collectors are made up of _____.

1. Metals
2. Metals with high thermal conductivity
3. Metals which can absorb heat easily and transfer the heat through conduction
4. All of the above

5 When a p-n junction is at equilibrium, its _____ is flat.

1. Band structure
2. Valence band at the junction
3. Conduction band at the junction
4. Fermi energy

6 Solar cell is a ____ while battery is a ____ based device.

1. Voltage source, Current source
2. Current source, voltage source
3. Voltage source, electron source
4. Ionic source, electronic source

7 Windmill's energy generation depends on the ____.

1. Volume of air going through it
2. Cube power of wind speed
3. Height of the windmill
4. All of the Above

What are the challenges with OTEC plant?

1. Pressure
2. Depth
3. Water Current
4. All of the Above

9 What kind of material/coatings is used for coating turbine blades?

1. Metallic coating
2. Composite
3. Ceramic
4. None

Battery and fuel cell ____.

1. Both are electrochemical devices
2. Both can be used for electric vehicles
3. Both a & b
4. None

11 Electrolyte used in fuel cell and Li ion battery are ____.

1. Same
2. Different
3. Electronic conductors
4. None

12 Which of the following are energy conversion devices

1. Battery
2. Fuel Cell
3. Capacitor
4. None

Q2. Solve the following.

- A) Write a short note on scale of quantities
- B) Discuss energy consumption by different sector

Q3. Solve the following.

- A) Write a short note on solar spectrum
- B) Enlist solar thermal devices & Explain solar flat plate collector

Q4. Solve Any Two of the following.

- A) Discuss performance & durability of solar photovoltaic device

- B) Explain with neat diagram Horizontal axis wind turbine (HAWT). 6
- C) Write a note on biomass. 6
- Q5. Solve Any Two of the following.
- A) Explain in detail geothermal energy with schematic diagram. 6
- B) Give detailed classification of battery & discuss Li-Ion Battery in detail. 6
- C) Write a short note on parameters which will affect performance of batteries 6
- Q6. Solve Any Two of the following.
- Give Types of Fuel cell & explain in detail alkali fuel Cell (AFC) 6
- Write a short note on fly wheel. 6
- What is Supercapacitor? Discuss in detail construction & Working of super capacitor 6

*** End ***